PREVENTION OF MOLECULAR WEIGHT REDUCTION OF THE POLYMER, IMPURITY FORMATION AND GELLING IN POLYMER COMPOSITIONS

RELATED APPLICATIONS

This application is a Continuation of application Ser. No. 10/894,956, filed Jul. 19, 2004, which claims benefit the of U.S. Provisional Application No. 60/488,573 filed Jul. 18, 2003. The text of application Ser. No. 10/894,956 and Application No. 60/488,573 is incorporated by reference in their entirety herewith.

FIELD OF THE INVENTION

The present invention relates to polymer compositions capable of providing sustained or extended drug release, methods of their preparation and use in various biological applications. Specifically, it relates to prevention or elimination of molecular weight reduction of the polymer in a polymer solution containing nucleophilic compound(s) or agent(s). It also relates to reduction or elimination of peptide polymer impurities in the polymer solutions. It further relates to GnRH analog and polymer containing solutions capable of forming stable, filter sterilizable, non-gelling solutions.

BACKGROUND OF THE INVENTION

Compositions containing biologically active agents in combination with biocompatible and biodegradable polymers are being increasingly used as drug delivery systems to provide sustained or delayed release of drugs. The compositions are available in various injectible depot forms including liquid forms, solid implants, microspheres, microcapsules and microparticles.

For example controlled release systems where the polymeric compositions are liquid forms or flowable delivery systems are described in U.S. Pat. Nos. 5,739,176, 4,938,763; 5,278,201, 5324519 and 5,278,202. The compositions described in these patents are administered to the body of a subject in a flowable state. Once in the body, the composition precipitates or coagulates to form a solid matrix or implant and the organic solvent in the composition dissipates or disperses into the aqueous or body fluid. Once the solid implant is formed, usually at the site of administration, the biologically active agent is released from the solid matrix by diffusion or dissolution from within the polymeric matrix and/or by the slow degradation of the polymeric matrix.

A variety of sustained release microspheres or microcapsules are also available or are being developed as delivery systems for the rapidly expanding class of peptide and non-peptide therapeutic or pharmacological agents. For example, sustained release microspheres or microcapsules are known in the administration of hormones, hormone analogs, antitumoral drugs, thioridazine, antipsychotic, and steroids where PLGA o PLA is the constitutive biodegradable polymer material. Further, for example, in recent years, a variety of injectable depot formulations in which a somatostatin analog (e.g., octreotide acetate) or a LHRH analog (e.g. leuprolide acetate) encapsulated in, and released slowly from, microspheres made of biodegradable polymers have been reported (U.S. Pat. Nos. 5,478,564, 5,540,973, 5,609,886, 5,876,761, 5,688,530, 4,652,441, 4,677,191, 4,917,893, 4,954,298, 5,330,767, 5,476,663, 5,575,987, 5,631,020, 5,631,021 and 5,716,640). Indeed, long acting injectable depot formulations of GnRH analogues (agonists and antagonists) are being used and/or

tested for the treatment of various pathological and physiological conditions in mammals, particularly in humans (Kostanski et al., 2001, *BMC Cancer,* 1:18-24). The treatments are for, among other things, the management of sex hormone-dependent diseases such as prostate cancer and endometriosis, for the induction of ovulation, and for the control of male fertility.

Thus, significant efforts are being made to maintain a steady release of medicinal drugs in animals by using compositions containing biodegradable biocompatible polymers. One obvious goal behind in all of these polymeric compositions is that the biologically active agent (e.g., a peptide or protein) of interest can be administered less frequently, sometimes at lower overall doses, than when formulated as a solution without the use of polymers in them. More importantly, it can justify commercial development of proteins that, for a variety of reasons, could not be marketed as simple solution formulations.

Despite the technological advances that were made in the area of injectable depot formulations to date, a number of quality concerns prevent their ready use in biological applications. These include reduction in molecular weight of the polymers in the polymer composition, de novo formation of conjugate substances (impurities), insolubility of the biologically active agents in solvents typically used in the polymer compositions and their propensity to form gels.

Molecular weight of the polymer material (e.g., PLGA/PLA matrix) is an important factor in designing sustained release formulations because drug release profile and the degradation rate of the polymer depend on molecular weight of the polymer in the final product. It has been reported that the molecular weight of the PLGA decreased in the microcapsules (microspheres) containing simple basic compounds such as thioridazine and ketotifen as free bases during their fabrication. Microcapsules fabricated from their pamoate salts did not produce much reduction in molecular weight (Maulding et al., 1986, Journal of Controlled Release, 3:103-117). U.S. Pat. No. 5,916,598 showed that the presence of benzyl alcohol, as the residual solvent in the microspheres reduced the shelf life of the product by molecular weight reduction and the patent provides a method to reduce the residual benzyl alcohol level. However, it is not always possible to remove the residual solvents from the microspheres.

U.S. Pat. No. 6,264,987 discloses that the simple nucleophilic compounds such as risperidone, naltrexone, and oxybutynin can degrade the PLGA variably depending upon the holding time and temperature of the dispersed phase solution. Methods provided to minimize the reduction in molecular weight are, lowering the hold time of the drug-polymer solution and the hold temperature. However, during the manufacturing of the sustained release products, it is very difficult to control the hold time of the drug-polymer solutions. Also, the hold time and its effect may depend upon the type, co-monomer ratio and co-monomer sequence of the polymer. Further, there could be unexpected delays in the aseptic processing sequence during the fabrication of microspheres, which could make the entire drug-polymer solution not usable. Lowering the hold temperature of the drug-polymer solution could result in drug crystallization or viscous polymer-drug solution. Higher viscosity solutions are difficult to sterile filter and often give rise to larger particles. Larger particles could pose syringeability problems.

The FDA and ICH guidelines on impurities in new drug substances suggest that any impurity (individual impurity) greater than 0.1% has to be reported, and any impurity greater than 0.15% has to be identified. If the impurity in a new drug is more than a given threshold level, those impurities should